

Exercise 4.5.3

Let $c \in (a, b)$. We will show that f is continuous at c . Since f is increasing, for every $x < c$, we have $f(x) \leq f(c)$. Let $L = \sup\{f(x) : a \leq x < c\}$. Then $L \leq f(c)$. We claim that $L = f(c)$. Assume for contradiction that $L < f(c)$. Then choose some number M such that $L < M < f(c)$. Since $f(a) \leq L < M < f(c)$, by the intermediate value property, there exists some $y \in [a, c]$ such that $f(y) = M$. But $y \neq c$, since $f(c) \neq M$. So, $y < c$. Therefore, by the definition of L as a supremum, we must have $f(y) \leq L$. This contradicts $f(y) = M > L$. Thus, $L = f(c)$.

Now, let $\epsilon > 0$. Since $L = f(c)$ and $L = \sup\{f(x) : a \leq x < c\}$, there exists some $x_0 \in [a, c)$ such that $f(c) - \epsilon < f(x_0) \leq f(c)$. Then, for any $x \in (x_0, c)$, since f is increasing, we have $f(c) - \epsilon < f(x_0) \leq f(x) \leq f(c)$. Thus, $|f(x) - f(c)| < \epsilon$. Therefore, $\lim_{x \rightarrow c^-} f(x) = f(c)$.

Similarly, $\lim_{x \rightarrow c^+} f(x) = f(c)$. Therefore, by Theorem 4.6.3, $\lim_{x \rightarrow c} f(x) = f(c)$, so f is continuous at c . Since $c \in (a, b)$ was arbitrary, f is continuous on (a, b) .

The endpoint cases follow similarly using one-sided limits. Therefore, f is continuous on $[a, b]$.

Exercise 4.5.4

If g is one-to-one on the interval A , then F will contain no points, and is therefore empty. So, we focus on the case where g is not one-to-one on the interval A . So, there exists a and b such that $a \neq b$ and $g(a) = g(b)$. Without loss of generality, let $a < b$. If g is constant on $[a, b]$, then every point $x \in [a, b]$ is also in F . Since $[a, b]$ is an interval with $a \neq b$, it is uncountable, so F is uncountable. Assume g is not constant. Then, there exists $x_0, x_1 \in [a, b]$ such that $g(x_0) \leq g(x) \leq g(x_1)$ for all $x \in [a, b]$, by the Extreme Value Theorem. Since g is not constant and $g(a) = g(b)$, at least one of x_0 or x_1 is not at the endpoints, as both the minimum and maximum values cannot equal $g(a)$. Let $c \in (a, b)$ be a point where g attains such a minimum or maximum.

First, suppose $g(c)$ is a maximum and $g(a) < g(c)$. The case where $g(c)$ is a minimum is analogous, using the interval $(g(c), g(a))$. For each $L \in (g(a), g(c))$, choose one such point $x_L \in (a, c)$ with $g(x_L) = L$, by the Intermediate Value Theorem. Also, since $g(b) = g(a) < L < g(c)$, the Intermediate Value Theorem gives some $y_L \in (c, b)$ such that $g(y_L) = L$. $x_L < c < y_L \implies x_L \neq y_L$, so $x_L \in F$. If $L_1 \neq L_2$, then $x_{L_1} \neq x_{L_2}$, since otherwise $L_1 = g(x_{L_1}) = g(x_{L_2}) = L_2$, which is impossible. Thus the set $\{x_L \mid L \in (g(a), g(c))\}$ is in one-to-one correspondence with $(g(a), g(c))$, which is an uncountable open interval. Therefore, since $\{x_L \mid L \in (g(a), g(c))\} \subseteq F$, F is uncountable.

Exercise 4.5.6

- (a) We can use $f(\frac{1}{2})$ to find a $x, y \in [0, 1]$ satisfying the condition. If $f(\frac{1}{2}) = f(0)$, we can use $x = 0$ and $y = \frac{1}{2}$. If $f(\frac{1}{2}) \neq f(0)$, assume $f(\frac{1}{2}) > f(0) = f(1)$ without loss of generality. The case $f(\frac{1}{2}) < f(0) = f(1)$ is analogous. Define $g(x) = f(x + \frac{1}{2}) - f(x)$ for $x \in [0, \frac{1}{2}]$. g is continuous, as it is the difference of two continuous functions. $g(0) = f(\frac{1}{2}) - f(0) > 0$ and $g(\frac{1}{2}) = f(1) - f(\frac{1}{2}) < 0$. By the Intermediate Value Theorem, there exists a $c \in (0, \frac{1}{2})$ such that $g(c) = 0$. Then, $0 = g(c) = f(c + \frac{1}{2}) - f(c) \implies f(c) = f(c + \frac{1}{2})$. Let $x = c$ and $y = c + \frac{1}{2}$. Therefore, there exists $x, y \in [0, 1]$ satisfying $|x - y| = |c - c - \frac{1}{2}| = \frac{1}{2}$ and $f(x) = f(y)$.
- (b) Define $g(x) = f(x + \frac{1}{n}) - f(x)$ for $x \in [0, \frac{n-1}{n}]$. g is continuous, as it is the difference of two continuous functions. Also,

$$\sum_{k=0}^{n-1} g\left(\frac{k}{n}\right)$$

$$\begin{aligned}
 &= \sum_{k=0}^{n-1} \left(f\left(\frac{k+1}{n}\right) - f\left(\frac{k}{n}\right) \right) \\
 &= \left(f\left(\frac{1}{n}\right) - f(0) \right) + \left(f\left(\frac{2}{n}\right) - f\left(\frac{1}{n}\right) \right) + \cdots + \left(f\left(\frac{n-1}{n}\right) - f\left(\frac{n-2}{n}\right) \right) + \left(f(1) - f\left(\frac{n-1}{n}\right) \right) \\
 &= f(1) - f(0) \\
 &= 0
 \end{aligned}$$

There are two cases: at least one $g\left(\frac{k}{n}\right) = 0$ for some $k = 0, 1, \dots, n-2, n-1$ or $g\left(\frac{k}{n}\right) \neq 0$ for all $k = 0, 1, \dots, n-2, n-1$. In the first case, it is clear that $f\left(\frac{k+1}{n}\right) - f\left(\frac{k}{n}\right) = 0$, so we can choose $x = \frac{k}{n}$ and $y = \frac{k+1}{n}$. Then, x, y satisfy $|x - y| = \frac{1}{n}$ and $f(x) = f(y)$. In the other case, some terms of g must be positive and others must be negative for the sum to be equal to zero. Then, g is positive at one point and negative at another point in $\left[0, \frac{n-1}{n}\right]$. So, by the Intermediate Value Theorem, there exists some $c \in \left[0, \frac{n-1}{n}\right]$ such that $g(c) = 0$. So, $g(c) = f\left(c + \frac{1}{n}\right) - f(c) = 0$. Let $x = c$ and $y = c + \frac{1}{n}$. Therefore, there exists $x, y \in [0, 1]$ satisfying $|x - y| = \left|c - c - \frac{1}{n}\right| = \frac{1}{n}$ and $f(x) = f(y)$.

(c) Let

$$f(x) = \begin{cases} -10x, & 0 \leq x \leq \frac{1}{5}, \\ 15x - 5, & \frac{1}{5} \leq x \leq \frac{2}{5}, \\ -10x + 5, & \frac{2}{5} \leq x \leq \frac{3}{5}, \\ 15x - 10, & \frac{3}{5} \leq x \leq \frac{4}{5}, \\ -10x + 10, & \frac{4}{5} \leq x \leq 1. \end{cases}$$

Also, let $g(x) = f\left(x + \frac{2}{5}\right) - f(x)$ for $x \in \left[0, \frac{3}{5}\right]$. We are looking for $g(c) = 0$ to satisfy $f\left(c + \frac{2}{5}\right) = f(c)$. Any two points $x, y \in [0, 1]$ with $|x - y| = \frac{2}{5}$ can be relabeled as c and $c + \frac{2}{5}$ for some $c \in \left[0, \frac{3}{5}\right]$. However, $g(x) = 1$ for all $x \in \left[0, \frac{3}{5}\right]$. So, there does not exist such a c . Therefore, there does not exist $x, y \in [0, 1]$ satisfying $|x - y| = \frac{2}{5}$ and $f(x) = f(y)$.

Exercise 4.5.7

To show that $f(x) = x$ for some $x \in [0, 1]$, we can show that $f(x) - x = 0$ for some $x \in [0, 1]$. Let $g(x) = f(x) - x$. $g(0) = f(0) - 0 = f(0) \geq 0$ and $g(1) = f(1) - 1 \leq 0$, since the range of f is contained in $[0, 1]$. If $g(0) = f(0) = 0$, we can choose $x = 0$ and have $f(x) = x$. If $g(1) = f(1) - 1 = 0$, then $f(1) = 1$, so we can choose $x = 1$ and have $f(x) = x$. In the case that neither of those is true, we have $g(0) > 0$ and $g(1) < 0$. Since g is the difference of continuous functions, it is also continuous. Then, by the Intermediate Value Theorem, there exists a $c \in (0, 1)$ such that $g(c) = 0$. Then, $g(c) = f(c) - c = 0$. So, $f(c) = c$. Therefore, we can choose $x = c$, and we have $f(x) = x$.

Exercise 4.6.4

First, we will prove the forward direction. Assume $\lim_{x \rightarrow a} f(x) = L$. We want to prove that $\lim_{x \rightarrow a^+} f(x) = L$ and $\lim_{x \rightarrow a^-} f(x) = L$. We know that for all $\epsilon > 0$, there exists a $\delta > 0$ such that $0 < |x - a| < \delta$ implies $|f(x) - L| < \epsilon$. If $0 < x - a < \delta$, then $0 < |x - a| < \delta$, so $|f(x) - L| < \epsilon$. Therefore, $\lim_{x \rightarrow a^+} f(x) = L$. Similarly, if $0 < a - x < \delta$, then $0 < |x - a| < \delta$, so $|f(x) - L| < \epsilon$. Therefore, $\lim_{x \rightarrow a^-} f(x) = L$.

Now, we will show the backward direction. Assume $\lim_{x \rightarrow a^+} f(x) = L$ and $\lim_{x \rightarrow a^-} f(x) = L$. We want to prove that $\lim_{x \rightarrow a} f(x) = L$. From the assumptions, we have two statements. For each $\epsilon > 0$, there exists a $\delta_1 > 0$ such that $0 < x - a < \delta_1$ implies $|f(x) - L| < \epsilon$. Also, for each $\epsilon > 0$, there exists a $\delta_2 > 0$ such that $0 < a - x < \delta_2$ implies $|f(x) - L| < \epsilon$. Choose $\delta = \min\{\delta_1, \delta_2\}$. Then, for each $\epsilon > 0$, $0 < x - a < \delta$ implies $|f(x) - L| < \epsilon$ and $0 < a - x < \delta$ implies $|f(x) - L| < \epsilon$. Combining the two inequalities, for each $\epsilon > 0$, $0 < |x - a| < \delta$ implies $|f(x) - L| < \epsilon$. Since we can always choose such a δ , the backward direction

is proved. So, $\lim_{x \rightarrow a} f(x) = L$ if and only if both the right and left-hand limits equal L .

Exercise 4.6.5

Let f be a monotone function with a discontinuity at c . We are going to prove that it must be a jump discontinuity. Assume that it is not a jump discontinuity. Since f is monotone, the one-sided limits $\lim_{x \rightarrow c^-} f(x)$ and $\lim_{x \rightarrow c^+} f(x)$ exist. Since there is not a jump discontinuity at c , $\lim_{x \rightarrow c^+} f(x) = \lim_{x \rightarrow c^-} f(x)$. First suppose f is monotone increasing. The monotone decreasing case is analogous. Let the left- and right-hand limits be equal to L . There are two cases: $f(c) = L$ or $f(c) \neq L$. If $f(c) = L$, then f is continuous at c , which is a contradiction to the original assumption that f has a discontinuity at c . For the case where $f(c) \neq L$, either $f(c) > L$ or $f(c) < L$. First suppose $f(c) > L$. The case $f(c) < L$ is analogous, using the left-hand limit. Choose $\epsilon = \frac{f(c) - L}{2} > 0$. Since $\lim_{x \rightarrow c^+} f(x) = L$, there exists $\delta > 0$ such that if $0 < x - c < \delta$, then $|f(x) - L| < \epsilon$. So, $f(x) < L + \epsilon = \frac{f(c) + L}{2} < f(c)$. Since $x > c$, this contradicts the fact that f is monotone increasing, so c must be a jump discontinuity. The monotone decreasing case is analogous.

Exercise 4.6.7

(a) $\mathbb{R}$ is F_σ because it can be written as the countable union of closed sets.

$$\mathbb{R} = \bigcup_{n=1}^{\infty} [-n, n]$$

$\mathbb{R} \setminus \{0\}$ is F_σ because it can be written as the countable union of closed sets.

$$\mathbb{R} \setminus \{0\} = \bigcup_{n=1}^{\infty} \left(\left[-n, -\frac{1}{n} \right] \cup \left[\frac{1}{n}, n \right] \right)$$

$\mathbb{Q}$ is F_σ because it can be written as the countable union of closed sets.

$$\mathbb{Q} = \bigcup_{q \in \mathbb{Q}} \{q\}$$

Since there are a countable number of rational numbers and singleton sets are closed, $\mathbb{Q}$ is F_σ .

(b) We want to show that $\mathbb{Z}^c$ is F_σ . By De Morgan's Laws, we can instead show that $\mathbb{Z}$ is G_δ . So, we need to write $\mathbb{Z}$ as a countable intersection of open sets. Let

$$O_n = \bigcup_{k \in \mathbb{Z}} \left(k - \frac{1}{n}, k + \frac{1}{n} \right)$$

Each O_n is open because it is a union of open intervals. Also, each O_n contains every integer, since for any $k \in \mathbb{Z}$, we have $k \in \left(k - \frac{1}{n}, k + \frac{1}{n} \right)$. So,

$$\mathbb{Z} \subseteq \bigcap_{n=1}^{\infty} O_n$$

Now, let $x \notin \mathbb{Z}$. Let z be a closest integer to x . Since $x \notin \mathbb{Z}$, we know $|x - z| > 0$. So, there exists some $N \in \mathbb{N}$ such that $\frac{1}{N} < |x - z|$. Then for every $k \in \mathbb{Z}$, we have $|x - k| \geq |x - z| > \frac{1}{N}$. So, $x \notin O_N$, so $x \notin \bigcap_{n=1}^{\infty} O_n$. Thus, $\bigcap_{n=1}^{\infty} O_n \subseteq \mathbb{Z}$. Therefore,

$$\mathbb{Z} = \bigcap_{n=1}^{\infty} O_n$$

Since $\mathbb{Z}$ is a countable intersection of open sets, $\mathbb{Z}$ is a G_δ set. Therefore, by De Morgan's Laws, $\mathbb{Z}^c$ is F_σ .

We want to show that $(0, 1]$ is F_σ . So, we need to write $(0, 1]$ as a countable union of closed sets. Let $F_n = [\frac{1}{n}, 1]$. Each F_n is closed.

$$\bigcup_{n=1}^{\infty} F_n = (0, 1]$$

So, $(0, 1]$ is F_σ .